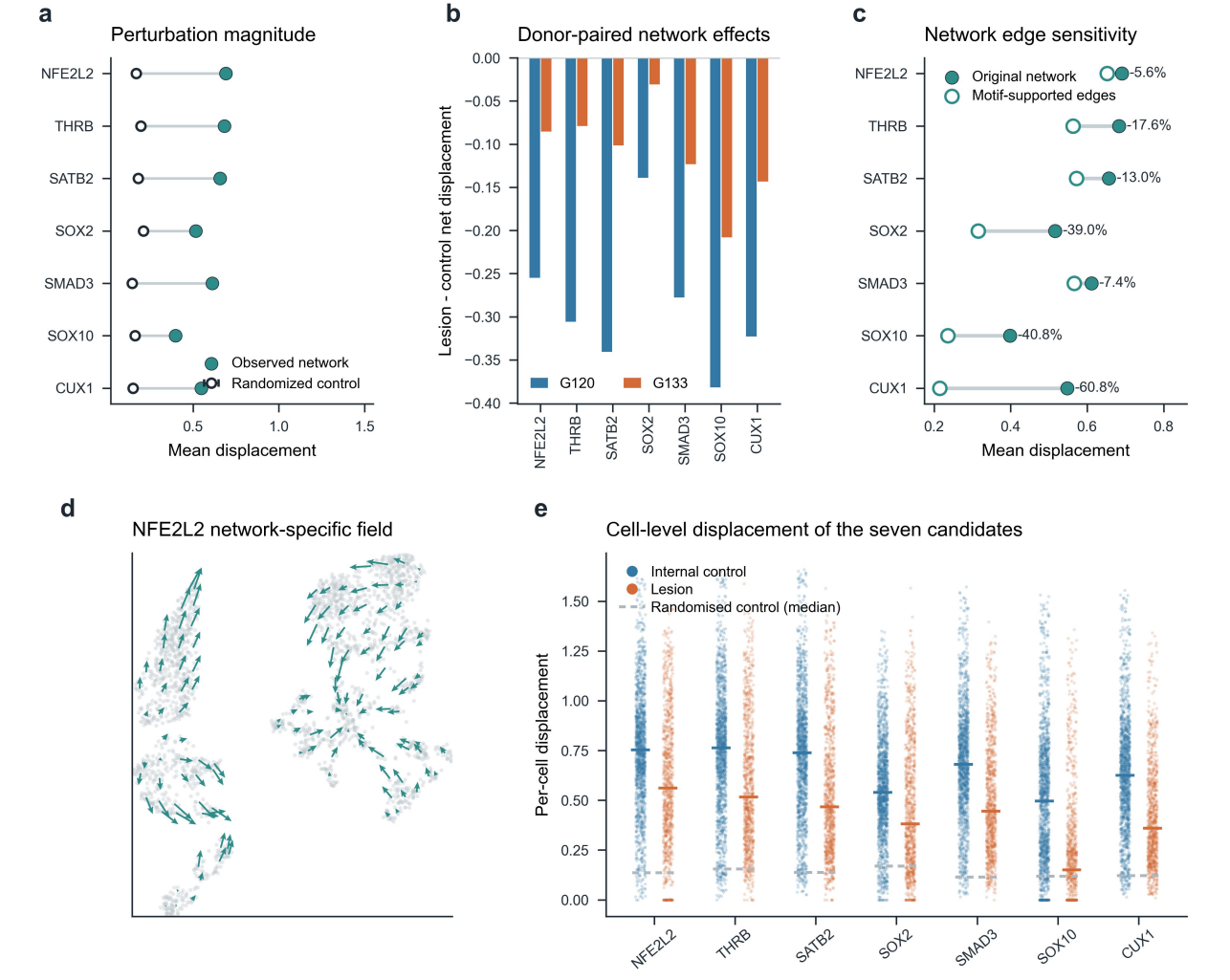

a
Perturbation magnitude
NFE2L2
THRB
SATB2
SOX2
SMAD3
SOX10
CUX1
Observed network
Randomized control
0.5
1.0
1.5
Mean displacement
b
Donor-paired network effects
Lesion - control net displacement
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
−0.30
−0.35
−0.40
G120
G133
NFE2L2
THRB
SATB2
SOX2
SMAD3
SOX10
CUX1
c
Network edge sensitivity
Original network
Motif-supported edges
-5.6%
-17.6%
-13.0%
-39.0%
-7.4%
-40.8%
-60.8%
0.2
0.4
0.6
0.8
Mean displacement
d
NFE2L2 network-specific field
e
Cell-level displacement of the seven candidates
Internal control
Lesion
Randomised control (median)
Per-cell displacement
1.50
1.25
1.00
0.75
0.50
0.25
0.00
NFE2L2
THRB
SATB2
SOX2
SMAD3
SOX10
CUX1